



Robotic disassembly of electric vehicle batteries: Technologies and opportunities

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ABSTRACT

The demand for electric vehicle (EV) battery services, such as repair, remanufacturing, and recycling, is rising as more EVs enter the market. Disassembly is an essential step in these services, but it is usually done manually, which is slow and costly. Therefore, robots have been proposed to disassemble batteries. This work provides an extensive overview of existing technologies in EV battery disassembly and potential robotics. The technologies investigated in this paper focus on how robots can enhance automation and flexibility. This paper summarises the characteristics and applicability of the potential technologies and highlights promising research directions to stimulate further exploration by researchers.

1. Introduction

The electric vehicle (EV) market has expanded rapidly in recent years, with battery demand reaching 550 GWh in 2022 (IEA(2023)). Most EV manufacturers provide an 8-year or 100,000-mile warranty on their batteries (WILSON, 2021). End-of-life (EoL) EV batteries contain toxic and corrosive metals and electrolytes, which can harm the environment and pose safety risks if not properly treated (Fan et al., 2020; Harper et al., 2019). Consequently, efficient and effective techniques for the EoL treatment of EV batteries are becoming increasingly important (Castelvecchi, 2021; Eddy, Pfeiffer, & van de Staaij, 2019; S. Park et al., 2016), and such EoL EV batteries possess high economic value when properly processed and utilised (Hua et al., 2020). The current landscape of EV battery waste management encompasses three main routes: remanufacturing, repurposing and recycling (Harper et al., 2019).

Remanufacturing restores EoL EV batteries to their original quality

through processes such as the replacement of damaged components. Repurposing involves reclaiming usable parts of the batteries for new applications (Schulz-Mönnighoff, Bey, Nørregaard, & Niero, 2021), while recycling aims to extract metals and chemicals from them (Hua et al., 2020; Martinez-Laserna et al., 2018; Standridge et al., 2014).

As illustrated in Fig. 1, disassembly is an essential step in all three routes (Harper et al., 2019). It demands high-levels dexterity and carries potential safety risks such as electric shock. Therefore, manual operators need adequate training to perform EV battery disassembly safely (Harper et al., 2019). However, the lack of staff qualified to operate EVs safely is a significant limitation to the sustainable utilisation of EV batteries (Vallance, 2023). A survey conducted by the Institute of the Motor Industry revealed that there were 52,000 qualified EV technicians in the UK, representing 22 % of all technicians in the country at the end of January of 2024 (IMI, 2024). Given the high labour costs associated with EV battery disassembly, there is a significant demand for the

Abbreviations: 2D, Two-dimensional; 3D, Three-dimensional; AI, Artificial intelligence; ANN, Artificial neural networks; APF, Artificial potential field; AR, Augmented reality; CNN, Convolutional neural networks; DFD, Design for disassembly; DL, Deep learning; DRL, Deep reinforcement learning; DSP, Disassembly sequence planning; DWA, Dynamic window approach; EEG, Electroencephalography; EMG, Electromyography; EoL, End-of-life; EV, Electric vehicle; GA, Genetic algorithm; GMM, Gaussian mixture model; GNN, Graph neural networks; HMM, Hidden Markov model; HRC, Human-robot collaboration; IMU, Inertial measurement unit; LfD, Learning from demonstration; LSTM, Long short-term memory; MCTS, Monte-Carlo tree search; ML, Machine learning; MR, Mixed reality; NA, Necessity of automation; PCB, Printed circuit board; PRM, Probabilistic roadmap method; PSO, Particle swarm optimisation; QR, Quick Response; RD, Robotic disassembly; RF, Radio frequency; RL, Reinforcement learning; RMP, Robotic motion planning; RNN, Recurrent neural networks; RRT, Rapidly-exploring random tree; RUL, Remaining useful life; SOC, State of charge; SOH, State of health; SVM, Support vector machine; TAA, Technical ability of automation; TAMP, Task and motion planning; UAV, Unmanned aerial vehicle; VR, Virtual reality.

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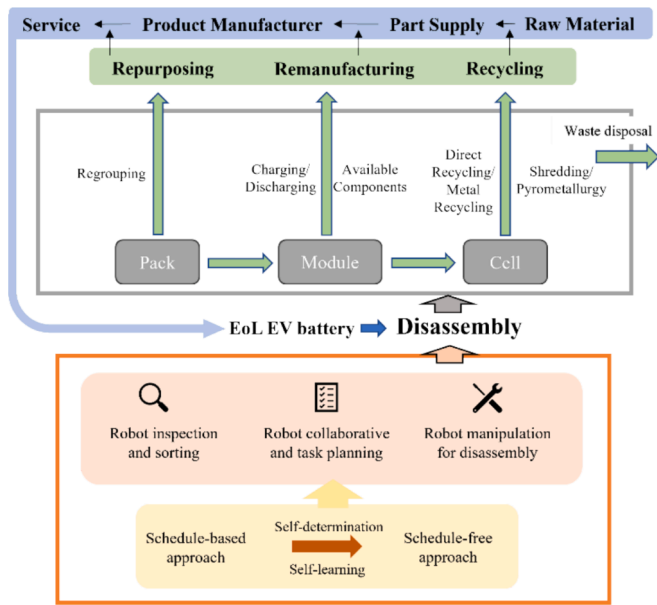


Fig. 1. Illustration of how disassembly helps with sustainable use of EV batteries and how the robot helps with disassembly. The upper shows the role of the disassembly operation in the EoL EV battery cycle; the lower shows the main role of robotics in EV battery disassembly, where the yellow block shows the development of key robot capabilities for disassembly. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

adoption of automation technologies in this sector.

Robots have been extensively utilised in various industrial automation applications, including assembly, welding and handling, and have shown potential in the area of disassembly (Jie Li, Barwood, & Rahimifard, 2018). As demonstrated in Fig. 1, robots can significantly contribute to almost all disassembly functions and processes (e.g. target detection, task planning, and manipulation). The aim of this paper is to review research on the robotic disassembly (RD) of EV batteries. Section 2 outlines the current challenges faced in the area. Section 3 examines the key robotic technologies that could be applied to the disassembly of EV batteries. Finally, Section 4 identifies potential research areas within this field.

2. Challenges in EV batteries disassembly

RD is a concept that originated in 1990 with a reported case of TV disassembly by robots (Scholz-Reiter, Scharke, & Hucht, 1999). Subsequent project by Apple, namely Liam and Daisy showcased the potential of disassembly robots for the material recycling of iPhones (Apple, 2018; Rujanavech et al., 2016).

Currently, the disassembly of EoL EV batteries relies heavily on manual labour. However, demonstrations of robot-based automated disassembly lines are being explored for broader industrial applications (L. Zhou, Garg, Zheng, Gao, & Oh, 2021). From a research standpoint, most existing RD systems are geared towards the pack-to-module disassembly level, where robots are capable of executing basic operations, such as removing the battery cover. Nevertheless, more complex and dexterous tasks, like unplugging and cell separation (Wegener, Andrew, Raatz, Dröder, & Herrmann, 2014; Wegener, Chen, Dietrich, Dröder, & Kara, 2015), are still performed by humans. Interestingly, the module-to-cell disassembly level has received little attentions in existing literature (Kaarlela, Villagrossi, Rastegarpanah, San-Miguel-Tello, & Pitkäaho, 2024). The literature shows that some complex disassembly tasks such as separating spot welds and removing wires still lack automated solutions. This leads to the module-to-cell disassembly level being rarely mentioned.

The use of robots in EV battery disassembly process is not widespread, mainly due to challenges that current robotic techniques are unable to overcome. The complexities of EV battery disassembly, as depicted in Fig. 2, pose significant obstacles. These challenges are summarised below.

- **High variations of design** – The absence of standardised design in battery packs leads to various manufacturers adopting different architectures, configurations, and materials. This diversity complicates the robotic disassembly process due to the need for customised approaches for each design (Castelvecchi, 2021).
- **Diverse disassembly depths** – Disassembly depth, which refers to the extent of disassembled a product undergoes. The depth of disassembly varies due to different economic objectives and types of products, necessitating flexible planning approaches to accommodate these variations (Giudice & Kassem, 2009). This diversity significantly impacts the strategy and planning of disassembly processes.
- **High variations of disassembly tasks** – RD of EV battery involves the removal of numerous joints and connectors, each requiring tailored RD strategies, control policies, and end effectors. Particularly challenging are joints like glued or welded connections in battery modules, which may need destructive operations potentially harmful to the modules and cells (Meng, Xu, Peng, Youcef-Toumi, & Li, 2022).
- **Uncertainties in battery condition** – EoL EV batteries present considerable uncertainties in their physical conditions, such as rust or loosening, adding complexity to the disassembly process.
- **Safety risks** – EoL EV batteries typically contain hazardous materials and can retain a high voltage, making manual disassembly a risky endeavour due to threats like electric shock and hazardous leaks.

The high variation is the primary challenge for RD of EV battery and makes it impractical to use pre-defined planning and routing strategies in disassembly. Furthermore, the high variability among EV batteries necessitates inspection as a critical element in disassembly.

Uncertainty regarding the condition of EV batteries introduces another significant challenge for RD. Uncertainties about the battery's state and potential damage to components means that the disassembly process is not simply the reverse of the assembly process. This adds considerable complexity to the modelling of RD processes and necessitates a higher degree of flexibility, adaptability, and autonomous decision-making in RD.

Safety risks also present a significant challenge in employing robots for EV disassembly. Certain disassembly tasks are too complex or costly to fully automate, leading to the possibility of human-robot collaboration (HRC) within the disassembly line. Such collaboration brings additional safety concerns, including the risk of human-robot collisions.

In summary, these challenges highlight the need for technological advancements in robotic inspection, planning, collaboration, and manipulation for EV battery disassembly systems. To overcome these challenges, highly automated and flexible RD systems are essential. These systems require not just traditional automation components like fixtures and sensors, but also intelligent decision-making capabilities.

3. Enabling techniques for the RD of EV batteries

According to (Wegener et al., 2015), current RD for EV batteries is centred around manual disassembly. For robotic manipulation, traditional programming approaches are employed to perform basic tasks. This RD approach, which we term the schedule-based approach, depends heavily on pre-programmed motion plans. Some researchers have explored integrating HRC, where humans manage complex tasks, and robots are deployed for the repetitive and simple tasks (Laili, Wang, Fang, & Pham, 2022). Despite these efforts, there remains a pressing need for more advanced solutions capable of adapting to diverse scenarios and conditions.

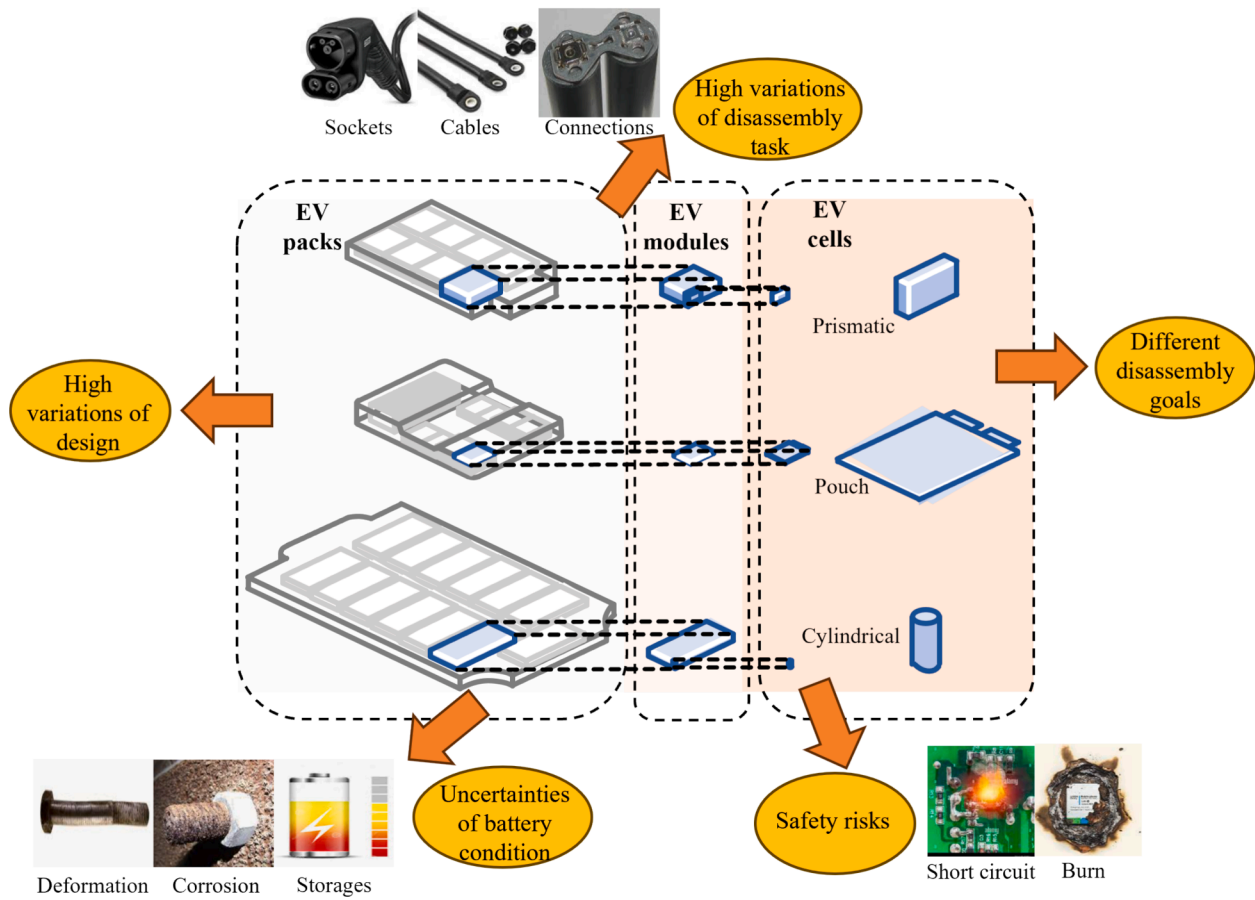


Fig. 2. Some types of EV battery structure and disassembly challenges (Castelvecchi, 2021). Three typical EV cell shapes are shown: prismatic, pouch, and cylindrical.

As shown in Fig. 3, an RD system for EV batteries includes several critical components and elements. The information of connections and constraints in EV battery is vital in assessing the feasibility of disassembly automation (Hellmuth, DiFilippo, & Jouaneh, 2021). This assessment covers two key aspects: the technical ability of automation (TAA) and the necessity of automation (NA). These criteria are helpful in

determining which segments of the disassembly process are suitable for automation, based on environmental, technological, and economic considerations (J Li, Barwood, & Rahimifard, 2019). The outcomes of this evaluation then guide the formulation of disassembly information for the RD system.

An RD system for EV batteries is generally comprised of two primary

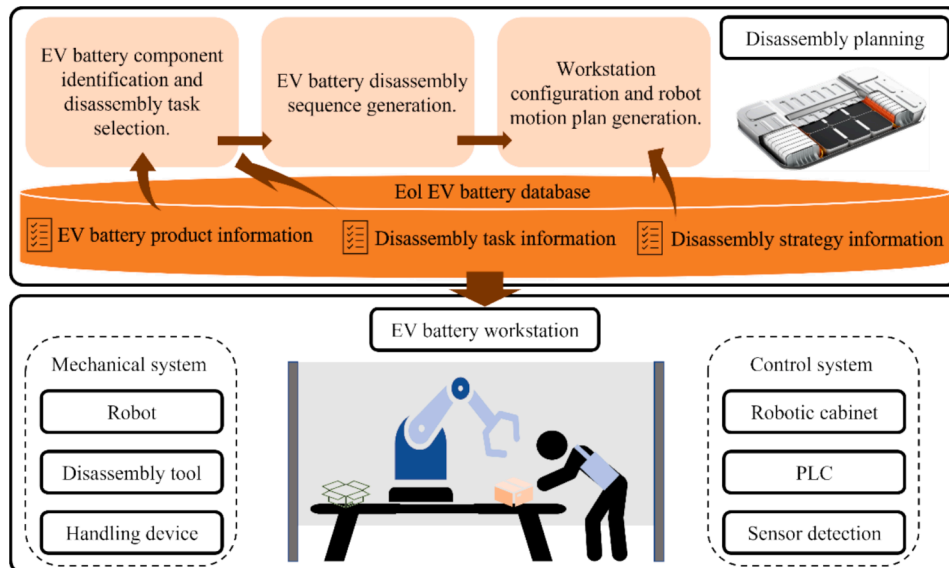


Fig. 3. A frame of disassembly workstations design.

components: an execution system and a disassembly planning system. The execution system, responsible for carrying out the physical disassembly tasks, consists of a mechanical system and a control system. The disassembly planning system, on the other hand, issues instructions to the execution system and undertakes two levels of tasks: disassembly sequence planning (DSP) and robotic motion planning (RMP). The disassembly schedule, which can be archived in a database (Baazouzi, Rist, Weeber, & Birke, 2021), should be flexible enough to accommodate both known and new EV batteries.

This section categorises various studies on RD technologies for EV batteries into three distinct themes:

- (1) **Robotic Inspection:** This crucial initial step, both before and during the disassembly of EV batteries, encompasses essential technologies such as sensors, robot vision systems, and procedures for sorting EV batteries. Robotic inspection is fundamental for identifying the state and type of the battery, which informs subsequent disassembly steps.
- (2) **Robotic Planning and Collaboration:** This theme focuses on planning, decision making, and collaboration techniques employed during robotic disassembly. Utilising data obtained from robotic inspection alongside human expertise, robots are programmed to plan and make decisions in both structured and dynamic disassembly environments. Covered topics include the planning of disassembly tasks, layout design of disassembly system, development of expert systems, RMP, etc. HRC is a key aspect under this theme, incorporating various techniques to enhance efficiency and safety.
- (3) **Robotic Disassembly Operation:** This theme addresses the actual execution of disassembly activities. It includes the use of flexible disassembly tools and robotic manipulations tailored to the disassembly process. Techniques specific to the disassembly of EV battery, such as grasping, separating, and removing components, are detailed in this section. This theme underscores the application of robotics in EV battery disassembly.

3.1. Literature collection

This review aims to summarise the current state and development of robotics in EV battery disassembly. The focus is on existing and potential robotic technologies applied to EV battery disassembly, which guided the keyword searches for relevant studies.

Several robotic technologies have been successfully demonstrated in EV battery disassembly. These findings show that robots can effectively perform EV battery disassembly, but significant gaps remain.

There is extensive literature of potential robotics-related technologies in EV battery disassembly. These technologies primarily originate from the fields of machine vision, mobile robotics, sensor design, manipulator design, HRC, and robot manipulation. The aim of these techniques is to improve the flexibility and automation of robotic disassembly to meet the challenges of EV battery disassembly. These techniques improve the capability of the robot at the perception level, decision level and execution level respectively.

The keywords are summarised in Table 1.

Additionally, there are technologies not commonly discussed in these popular robotics areas but are still important for EV battery disassembly. These technologies will be discussed in Section 4.

3.2. Robotic inspection in EV battery disassembly

Robotic inspection is critical to the EV battery disassembly. The classification and inspection of EV batteries before disassembly are essential for determining the appropriate final depth of disassembly (Kaarlela et al., 2024). Inspection is also crucial for safety monitoring and assessing the quality of disassembled products (Klohs et al., 2023).

Table 1

The taxonomy of technologies on RD of EV battery.

Field	Sub field	Keywords	Ref.
Existing robotic EV battery disassembly technologies	Review	EV battery robotic disassembly	(Kaarlela et al., 2024; Klohs, Offermanns, Heimes, & Kampker, 2023; Laili et al., 2022; Martinez-Laserna et al., 2018; Meng et al., 2022; Vongbunyong & Chen, 2015; L. Zhou et al., 2021)
	Robotic disassembly solution	Robotic disassembly platform	(Herrmann, Raatz, Mennenga, Schmitt, & Andrew, 2012; Kay, Farhad, Mahajan, Esmaceli, & Hashemi, 2022; Qu et al., 2024; Tan, Chin, Garg, & Gao, 2021; Wegener et al., 2014)
Potential robotic technologies for EV battery disassembly	Perception level	EV battery DSP and optimisation	(Baazouzi et al., 2021; M. Choux, Bigorra, & Tyapin, 2021)
		Machine vision	(Bang, Lee, & Kang, 2022)
	Decision level	Detection	(Cha, You, & Choi, 2016; Sampath, Yin, Tham, Chen, & Zhang, 2024)
		Robotic sensor DSP	(Gil et al., 2007)
		RMP	(Z. Zhou et al., 2019)
		Trajectory planning	(Mohanand & Salgoankar, 2018)
		Expert system HRC	(Gasparetto, Boscariol, Lanzutti, & Vidoni, 2015)
		Disassembly HRC	(Cakir & Cavdar, 2006)
	Execution level	Teleoperation	(Hjorth & Chrysostomou, 2022)
Disassembly tool		(Moniruzzaman, Rassau, Chai, & Islam, 2021)	
Robotic manipulator		(P. Schumacher & Jouaneh, 2013)	
Robot programming		(Bauer, Bauer, Lakshminpathy, & Pollard, 2022)	
	Robot learning	(Heimann & Guhl, 2020)	
	Compliance control	(Z. Liu, Liu, Xu, Wang, & Zhou, 2022)	
	Control method	(M. Schumacher, Wojtusich, Beckerle, & Von Stryk, 2019)	
		(Villani & De Schutter, 2016)	

3.2.1. Pre-processing in EV battery disassembly

The pre-processing stage involves identifying the model of EV batteries and assessing their conditions (Meng et al., 2022). Critical information such as the state of health (SOH), state of charge (SOC), and remaining useful life (RUL) (Lipu et al., 2018) are important metrics for the safety evaluation of EV batteries, and this information is also critical for EV battery disassembly. Vision systems can enhance EV battery disassembly by providing detailed information about components. Additionally, Quick Response (QR) codes and labels can help the vision system quickly retrieve necessary information (Meng et al., 2022).

Given the high variations and uncertainties of EV batteries, the difficulty of pre-processing inspection significantly increases (Hellmuth et al., 2021). There is a lack of comprehensive technical equipment to perform a thorough inspection of EoL EV batteries (Meng et al., 2022). To address these gaps, advanced intelligence methods can be proposed. Intelligent labelling technologies can be used to obtain EoL information,

providing contactless IDs that streamline the tracking and tracing of EV batteries (Wessel, Turetskyy, Wojahn, Hermann, & Thiede, 2020). Vision system and intelligent labelling technologies can be integrated, with labels offering service life data and vision systems assessing EoL conditions (Chu et al., 2018). A comparison of these two technologies is presented in Table 2.

3.2.2. EV battery components inspection

Component inspection is a crucial factor in assessing the overall condition of the EV battery, which influences the decision on whether to disassemble it. Unfortunately, the existing inspection and testing techniques are often inadequate in delivering an accurate and thorough assessment of individual cells and components (Meng et al.). Given that some EV batteries comprise numerous modules and a layered structure, the process of inspection and testing becomes repetitive. Here, industrial robots offer more efficient solutions for inspection.

Various non-destructive methods, such as electrochemical impedance spectroscopy, ultrasonic testing, and X-ray CT inspection (Wolf, Lafarge, & Brosius, 2019), are utilised not only to examine battery cells but also to assess other components. For example, robotic inspection can investigate the quality of welding joints in the battery, electrode substrates, and components of the battery pack (A. Sharma, 2023).

A machine vision-based inspection method is employed to detect surface damages on mechanical and electronic components and to gather additional information for subsequent disassembly (Cha et al., 2016). This method typically involves image acquisition, processing and recognition. Additionally, laser ultrasonic technology (Choi, Liu, Yi, Sampath, & Sohn, 2023) can reveal more details regarding disassembly uncertainties.

3.2.3. Sensors and techniques in EV battery disassembly process inspection

The inspection before RD, which includes the assessing the battery's status and the conditions of its part, provides a solid foundation for planning RD efforts. However, for a more flexible and automated RD process, in-process detection and monitoring during disassembly is also important (Klohs et al., 2023). For example, a policy for automatic threaded fastener removal has been suggested in (R. Li et al., 2020), relying on the robot's force sensor in the robot and the vision system for calibration.

Sensors plays a crucial role in the successful implementation of RD (Zou, Ge, Wang, Cretu, & Li, 2017). Often, a vision system is needed to identify and recognise components for the purpose of process monitoring (Vongbunyong & Chen, 2015). Additionally, other sensors, such as torque and force sensors, offer valuable data for both monitoring and controlling the disassembly process (D. Xu, Loeb, & Fishel, 2013). The application of tactile sensors in disassembly is gaining traction as well, as they can aid in developing more precise motion plans (Laili et al., 2022).

Sensors play a crucial role in providing data to the RD system. For example, visual sensors can be used to build CAD model for disassembly planning. Force and tactile sensors can indicate the contact in disassembly. Table 3 provides an overview of using sensors in relevant research.

Vision systems are widely considered the most effective detection method due to the flexibility and versatility (Meng et al., 2022).

Table 2

Two typical technologies for EV battery pre-processing.

Type	Advantages	Disadvantages	Ref.
Intelligent labelling	• Contactless unique ID • Efficiency	• Additional reading equipment is required • Need for uniform standards	(Wessel et al., 2020)
Machine vision	• Only image input is needed • Flexibility	• Performance depends on imaging conditions, sufficient training data and EoL battery uncertainty	(Chu et al., 2018)

Table 3

EV battery disassembly process data characterisation.

Data	Application	Ref.
2D-RGB image	Screw and fastener modelling	(Foo, Kara, & Pagnucco, 2021; X. Li et al., 2021) (Bai, Yang, Xu, & Yao, 2021; Brogan, DiFilippo, & Jouaneh, 2021; Mangold, Steiner, Friedmann, & Fleischer, 2022)
	Laptop part modelling	(Su, Hu, & Lou, 2021)
	Circuit board component modelling	(Jun Chen, Bao, & Pan, 2022)
	EV battery modelling	(Martin Choux, Marti Bigorra, & Tyapin, 2021)
3D Point Cloud	Instance segmentation for motors	(Wu et al., 2023)
	Instance segmentation and modelling for EV battery	(Bilal, Tyapin, & Choux, 2023; Brådlund, Choux, & Cenkramaddi, 2021)
Force/Torque data	Peg-in-hole disassembly	(Yongquan Zhang et al., 2019)
	Dual peg-in-hole disassembly	(Goli et al., 2024)
	Robotic unplugging operation in interference fit peg-in-hole	(S. Xu, Pham, & Su, 2024)
	Snap-fit disassembly and assembly	(X.-L. Guo & Sun, 2023)
Haptic data	Peg-in-hole insertion process	(Kim & Rodriguez, 2022)

Traditional image processing techniques often rely on thresholding (Mardia & Hainsworth, 1988) or template matching for classification (Hashemi et al., 2016). Given the complexity and high variability of EV battery parts, classification is typically conducted using Artificial Intelligence (AI) methods such as Artificial Neural Networks (ANN) and Support Vector Machine (SVM).

In recent developments, Deep learning (DL) coupled with machine vision has shown significant capabilities in inspection tasks (Z.-Q. Zhao, Zheng, Xu, & Wu, 2019). DL, a subset of Machine Learning (ML), utilises ANN with representation learning. Popular models for DL include Convolutional Neural Networks (CNN) (Yildiz & Wörgötter, 2019), Recurrent Neural Networks (RNN) (Khan, Li, & Luo, 2019) and Graph Neural Networks (GNN) (Kuo, Ashmore, Huggins, & Kira, 2019). Choosing the appropriate model depends on the specific issue being addressed, with the suitability largely influenced by data's dimensionality (Z. Liu et al., 2022).

A key advantage of DL is its ability to learn, allowing it to conduct both feature extraction and classification labelling in complex environments. Such intelligence algorithms have proven effective in various disassembly scenarios. For instance, a trained CNN can detect small components on a printed circuit board (PCB) (Kuo et al., 2019). Additionally, the development of a generic screw detection CNN has been noteworthy (Yildiz & Wörgötter, 2019).

Incorporating more dimensional sensors enhances the RD's sensing abilities for EV batteries, increasing flexibility. However, integrating numerous sensors can also introduce redundant information and elevate uncertainty. Consequently, data fusion from multiple sensors can process reliable information by algorithm to improve accuracy.

The fusion of multiple sensors enhances both detection capabilities and efficiency. For example, Vadakkepat et al. introduced a multimodal detection framework that combines sonar sensors, haptic sensors, and vision systems, significantly improving detection efficiency and tracking performance (Vadakkepat, Lim, De Silva, Jing, & Ling, 2008). Multi-sensor fusion has been beneficial in RD as well, as demonstrated by Calandra et al. (Calandra et al., 2018), where a deep multimodal convolutional network predicted the efficacy of grasps using raw haptic-visual data, which could be fine-tuned for optimised grasping actions.

3.3. Robotic planning and collaborative in EV battery disassembly

EV batteries possess a complex structure, which complicates both robot inspection and robotic planning. Fig. 4 illustrates an example RD workstation, highlighting key planning considerations (Vongbunyong & Chen, 2015). The unpredictability of EoL conditions necessitates the use of knowledge databases with reasoning capabilities and advanced automated planning methods for EV battery disassembly (Poschmann, Brüggemann, & Goldmann, 2021).

Given the complexities involved in EV battery disassembly, HRC has become a favoured strategy. Ensuring operational safety requires meticulous planning in the robotic disassembly. This includes addressing potential collisions between humans and robots and improving the robot's adaptability to minimise safety risks. In this scenario, teleoperation emerges as an innovative approach to HRC, aimed at enhancing the safety of the disassembly process (Meng et al., 2022).

3.3.1. DSP

DSP focuses on determining the order in which components are removed from an EV battery during the disassembly process. It involves decision-making and planning amidst significant complexity and

uncertainty and aims to achieve a variety of goals such as costs reduction, increased speed, and enhanced efficiency (Z. Zhou et al., 2019). In certain situations, DSP may also need to be adaptive or 'online' (X. Zhang, Fang, Liu, & Yazdani, 2024). These methods aim to adapt to the uncertainties of the disassembly state and adjust the disassembly sequence in real-time during the disassembly process.

Typically, DSP begins by constructing a mathematical model that represents the assembly structure of components within a product. Following this, a planning method is employed to develop disassembly plans that meet specific objectives, ranging from retrieving selected components to achieving complete disassembly (Z. Zhou et al., 2019).

Given the considerable complexity and variability in EV batteries disassembly, DSP often presents challenges and supports diverse disassembly objectives (Z. Zhou et al., 2019). These varied objectives indicate that DSP can be optimised in multiple directions. For example, if reducing disassembly time is the objective (JiaZhao Chen, Zhang, & Liao, 2011), the optimisation goal could be identifying a disassembly stopping point (or level) at which environmental, physical, and economic costs are minimised (Smith, Hsu, & Smith, 2016). For larger batteries, the travel time of robotic tools between tasks might also factor into the overall disassembly time (Azab, Ziout, & ElMaraghy, 2011).

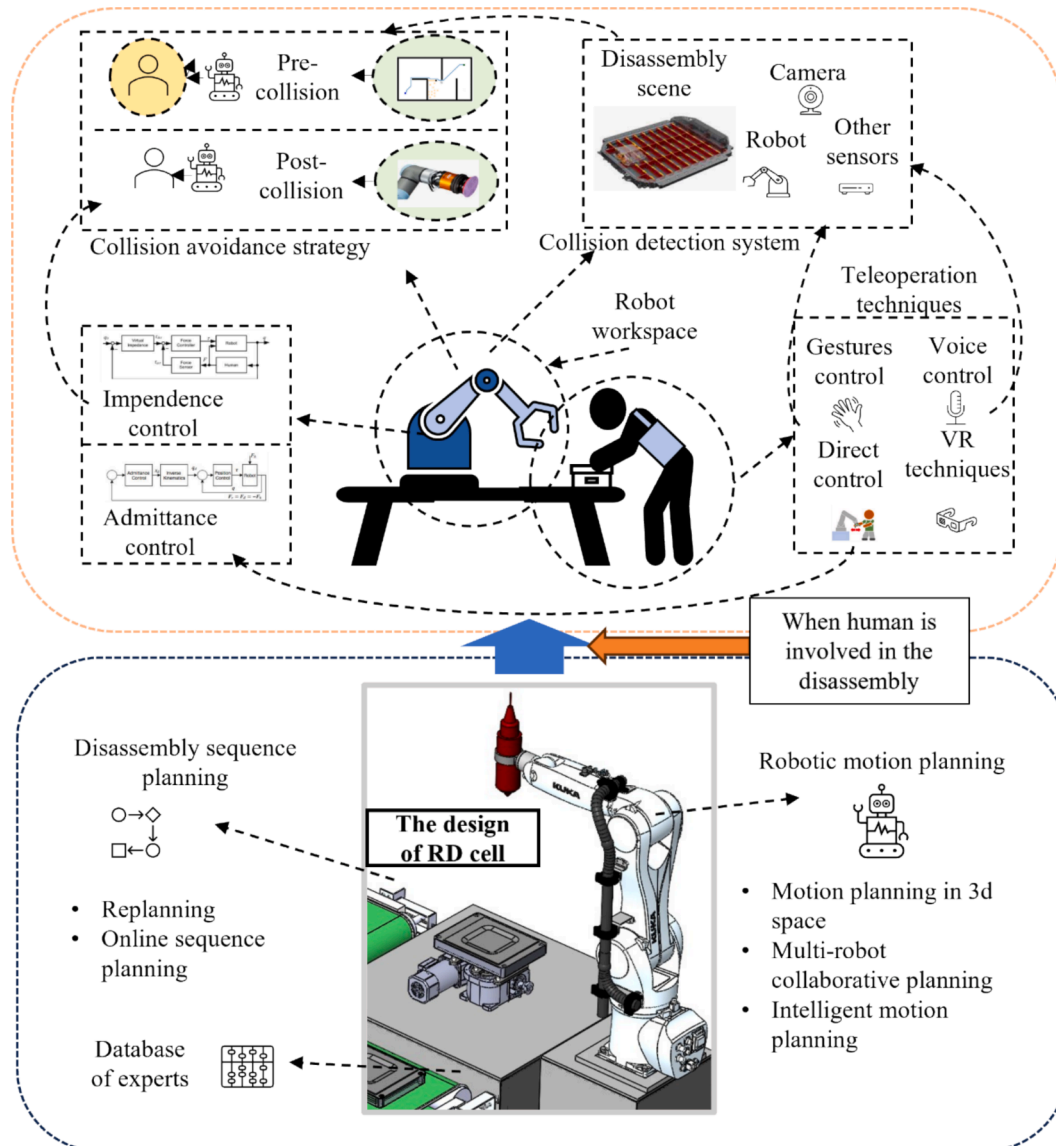


Fig. 4. An example RD workstation.

Optimisation goals can also focus on safety and reliability, particularly influenced by the choice between non-destructive (slower but safer) and destructive (faster but riskier) disassembly methods.

The presence of worn, corroded, or even missing components in EV batteries introduces uncertainties that can lead to inefficient and inaccurate disassembly plans (Laili, Tao, Pham, Wang, & Zhang, 2019). To mitigate these uncertainties, heuristics methods have been employed to allow a DSP system to adjust or replan in real-time. This capability for online replanning can be further enhanced through AI algorithms (Laili et al., 2019). Many researchers have explored heuristics based on evolutionary or iterative computational algorithms, such as genetic algorithm (GA) and swarm optimisation algorithms, to address the disassembly sequence problem (Feng et al., 2019; J. Liu et al., 2020).

Robotic learning techniques prove particularly effective in managing stochastic issues and replanning needs. For example, DSP challenges can be addressed through Reinforcement learning (RL) (Švaco, Jerbić, Polančec, & Šuligoj, 2018) or network prediction model (Lee, Liu, Behdad, Liang, & Zheng, 2022).

Some examples of DSP reported in the literature can be found in Table 4.

3.3.2. RMP

RMP represents a critical topic of RD. As illustrated in Fig. 5, components within EV are usually tightly assembled, requiring a formulated motion plan to avoid collisions and guarantee efficient disassembly.

To implement the RD of EV batteries, a comprehensive map detailing obstacles is essential for devising a robotic motion planning (Mohanan & Salgoankar, 2018). The map can be autonomously produced through planning algorithms, examples of which are delineated in Table 5. While these algorithms are typically employed offline, they may encounter challenges during real-time application (Mohanan & Salgoankar, 2018).

The path planning algorithms referenced previously excel in two-dimensional (2D) scenarios; however, RD of EV batteries often involves three-dimensional (3D) considerations, where the complexity of path planning significantly escalates (Yang et al., 2016). Although sampling-based method are applicable to 3D path planning (Yang et al., 2016), their success largely relies on the quality of the 3D models.

Trajectory planning is also a crucial part of motion planning. Trajectory planning algorithms assign time information to a given geometric path, affecting not only the kinematic properties of the movement but also the dynamic properties (Gasparetto et al., 2015). For a selected path, the position, speed, and acceleration of path points can be computed to generate the entire robot motion (Gasparetto et al., 2015). Some RMP works also discuss the joint space trajectory planning using polynomial interpolation and spline interpolation (Saramago and Steffen, 1998), which are simple and ensure smooth changes in velocity and acceleration during motion.

Trajectories can also be optimised using high-order polynomial parameterisation methods (Dinçer & Çevik, 2019). Some algorithms can

obtain optimal trajectories that satisfy kinematic and dynamic constraints, such as GA (L. Tian & Collins, 2004) and PSO algorithm (Du & Chen, 2022).

Illustrated in Fig. 6 is the feasibility of deploying multiple robots to operate simultaneously within the same workspace. This arrangement necessitates more intricate motion planning to accommodate multi-robot collaboration.

Centralised and decoupled are the two main approaches for multi-robot motion planning (Madridano, Al-Kaff, Martín, & De La Escalera, 2021). The centralised approach considers all robots as parts of a single integrated mechanical system, which aligns the problem closely with a traditional RMP scenario (Jose & Pratihari, 2016). However, this approach introduces a high number of degrees of freedom within the integrated mechanical system, significantly complicating motion planning (e.g. angle, speed and acceleration) (LaValle & Hutchinson, 1998).

In the decoupled approach, planning is typically segmented into two stages. The initial phase focus on determining collision-free paths for each robot independently of the others. The subsequent phase adjusts the robots' speeds to avoid collisions. However, this method does not guarantee that a solution will be found whenever one is available (Sanchez & Latombe, 2002). Some studies have blended elements of both planning approaches within the decoupled framework through an incremental coordination strategy (Saha & Isto, 2006).

The interplay between DSP and RMP is significant. DSP can influence the accessibility of components and the available manipulation space, thereby impacting RMP. Conversely, RMP can affect the duration of disassembly operations, thus influencing DSP outcomes. Consequently, some research has summarised the combination of DSP and RMP challenges under the umbrella of Task and Motion Planning (TAMP) (Deshpande, Kaelbling, & Lozano-Perez, 2020), emphasising the importance of linking high-level DSP considerations with low-level RMP issues. Addressing the TAMP problem requires identifying and analysing parameters that bridge the gap between DSP and RMP (Garrett et al., 2021).

Traditional motion planning algorithms work well in known environments but struggle with dynamic obstacle avoidance in unknown, high-dimensional maps. Intelligent planning methods can fuse information from various sources, such as target information, sensor information, and robot attributes, to plan optimal robot paths. In dynamic programming, the Q-learning algorithm is commonly used to search for the maximum reward value and choose the optimal path. Some studies have proposed improved Q-learning algorithms to reduce search complexity and feedback time (Jaradat, Al-Rousan, & Qadan, 2011). Alternatively, different search strategies can be used to reduce the specific search area and decrease algorithm complexity (Konar, Chakraborty, Singh, Jain, & Nagar, 2013).

Integrating image processing technology and dynamic tracking, visual servo systems can monitor demonstrated trajectories and execute dynamic planning based on RL outcomes (Miljković, Mitić, Lazarević, &

Table 4
Some examples of DSP case.

Product	Disassembly model	Disassembly objective	Optimisation algorithm	Ref.
Gearbox	Multi-layer matrix representation based on product bill of materials	Selective disassembly for maintenance	Integration of the ant colony method and the sequence search	(Y. Luo, Peng, & Gu, 2016)
Computer case	The disassembly stability matrix, the fastener precedence matrix, the fastener-component matrix, and the component motion precedence matrix	Human-robot collaborative disassembly considering disassembly time, cost, and difficulty	Modified discrete Bees algorithm based on Pareto	(W. Xu et al., 2020)
Engine block	One-way graph of the original disassembly structure of the product	Optimal disassembly time for complete disassembly of components	Multi-objective GA	(Go, Wahab, Rahman, Ramli, & Hussain, 2012)
Drive assembly	An inverted tree disassembly sequence structure graph and constraint matrix	Time-optimised partial disassembly of multiple target parts	A GA searches disassembled sequence graphs	(Smith, Smith, & Chen, 2012)
Hydraulic pump	A disassembly hybrid graph model	Reduce total disassembly time considering both disassembly tool changes and disassembly direction changes	A novel hybrid approach with a chaotic mapping-based hybrid algorithm of artificial fish swarm algorithm and GA	(J. Guo, Zhong, Li, Du, & Guo, 2019)



Fig. 5. An EV battery disassembly example (2017 Chevrolet Bolt EV 350.4 V Li-Ion Battery).

Table 5
Typical RMP methods.

RMP methods		Example	Advantages	Disadvantages	Ref.
Classical methods	Sampling based	Rapidly-exploring random tree (RRT), Probabilistic Roadmap Method (PRM)	- The algorithm structure is simple and easy to implement. - Variations of these methods (such as RRT* and PRM*) can lead to results that approach the optimal	Generate the sub-optimal path in the fixed workspace.	(Alarabi, Luo, & Santora, 2022; Noreen, Khan, & Habib, 2016)
	Reaction based	Artificial potential field (APF), Dynamic Window Approach (DWA)	This method has a good performance in dynamic obstacles.	It is easy to trap in the local minimum.	(Montiel, Sepúlveda, & Orozco-Rosas, 2015; H. Wang, Ma, & Zhu, 2022)
Learning based methods	Bio-inspired heuristic	GA, Particle Swarm Optimisation (PSO)	Path optimisation works better.	Less efficient in dynamic environments.	(Kucuk, 2017; Tarokh & Zhang, 2014)
		SVM	- Complex planning tasks can be accomplished with large amounts of data. - Its multi-class variant is effective for practical tasks.	SVM is used for 2-class classification which is not suitable for practical tasks,	(J. Tian, Gao, & Lu, 2007)
		Long Short-Term Memory (LSTM)	- Decisions for motion planning are made by executing the trained network. - It is more suitable for complex tasks.	The neural network training process requires a lot of data	(Inoue, Yamashita, & Nishida, 2019)
		Monte-Carlo Tree Search (MCTS)	- Can be more adaptable to dynamic environments. - MCTS is suitable for discrete action tasks.	It may not work in some practical problems.	(Paxton, Raman, Hager, & Kobilarov, 2017)

Babić, 2013).

3.3.3. Expert reasoning and database

In certain cases, disassembly tasks can be orchestrated using a knowledge-based expert system (Leo Kumar, 2019). This system is a computer program equipped with the capability to make decisions and solve problems independently. Based on the complexity of task, it can perform functions such as perception, reasoning, and decision-making (X. Xu, Wang, & Newman, 2011). Research on DSP based on knowledge graphs for reasoning has already been reported in the field of EV battery disassembly (Ding, Xu, Liu, Zhou, & Pham, 2019), but a complete expert system has yet to be developed. Fig. 7 shows a knowledge-based expert system, which comprises three core elements: a database, a knowledge base, and an interface engine. Databases store factual or heuristic data (often represented mathematically) pertinent to a specific problem domain (Cakir & Cavdar, 2006). This expert system is used in the field of metal cutting and provides process guidance. Incorporating a database into an expert system offers a tool to deal with uncertainties in EV battery disassembly processes. However, such systems are confined to the extent of their pre-existing knowledge and may not be effective in

cases with substantial knowledge gaps, like unknown battery models.

3.3.4. HRC in RD

When focusing on EV battery disassembly, many researchers have sought to enhance the design of the HRC system. To improve the disassembly sequence within HRC systems, optimisation algorithms such as the discrete bee algorithm (W. Xu et al., 2020) and the enhanced discrete bee algorithm (J. Liu et al., 2018) have been introduced. Given the human element in HRC systems, programming techniques have progressed from static offline programming to more flexible methods. A notable approach is demonstration-based programming, which facilitates the visualisation of disassembly frameworks (Schou, Andersen, Chrysostomou, Bøgh, & Madsen, 2018). Alternatively, robots can be adaptively modelled and programmed by utilising sensors to replicate operator disassembly skills (Vongbunyong, Vongseela, & Sreerattanaaporn, 2017).

Safety is a critical requirement for HRC system design. Multimodal sensors are employed to assess and predict the state of both artificial and disassembly systems, along with the disassembly environment (Q. Liu et al., 2019), ensuring a secure disassembly process. The HRC system

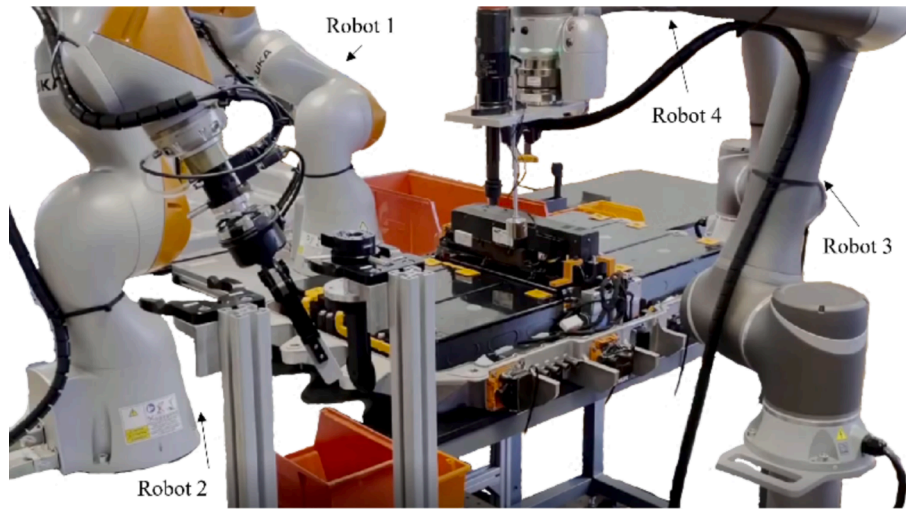


Fig. 6. A scene of a multi-robot disassembly cell for an EV battery (Qu et al., 2024).

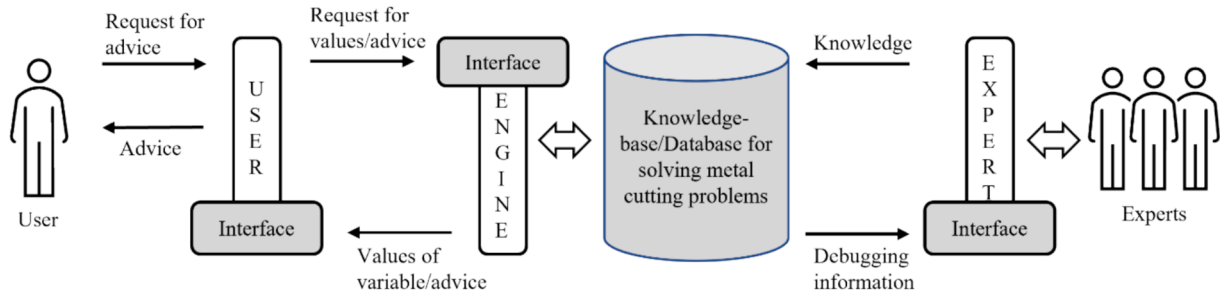


Fig. 7. A knowledge-based expert system in metal cutting (Cakir & Cavdar, 2006).

must integrate all equipment conditions and process data to execute a comprehensive risk evaluation. Through predictive protection and maintenance, potential failure in the disassembly system can be prevented. Environmental factors such as temperature, smoke, humidity, air purity, and quality ought to be constantly monitored to prevent the release of hazardous materials and avert perilous incidents. Additionally, psychological and cognitive ergonomics play a crucial role in HRC disassembly (Gualtieri, Rauch, & Vidoni, 2021). For example, the level of trust that human operator place in their robotic colleagues can significantly influence their collaborative efficiency (Akash, Hu, Jain, & Reid, 2018).

HRC has seen application in various disassembly scenarios. For example, Huang et al. investigated the disassembly of car water pump and turbo charger with robots assisting humans in the process of extracting components (Huang et al., 2021; Huang et al., 2020). Moreover, the removal of screws constitutes an important activity at HRC workstations (Chen et al., 2014; Li et al., 2020).

3.3.5. Robot techniques for HRC

In HRC systems, the potential for collision between humans and robots exists. To manage collisions effectively, robots are required to detect, recognise, and respond to both potential and actual collisions (Hjorth & Chrysostomou, 2022). The RMP method employed in schedule-based disassembly, might not be entirely suitable for dynamic disassembly environments characteristic of HRC (Mohanani & Salgoankar, 2018).

To prevent collisions in HRC, a pre-collision control strategy utilises sensors, including cameras and lasers. These sensors monitor the positions and poses of the human operators, enabling the motion planner to formulate collision avoidance strategies (Safeea, Neto, & Béarée, 2019). Enhancements in sensor efficiency and adoption of reactive collision

avoidance measures can further refine these strategies (Landi, Ferraguti, Costi, Bonfé, & Secchi, 2019). Vision systems, such as multiple KinectV2 cameras, are frequently deployed to identify dynamic obstacles (Tashtoush et al., 2021), while techniques like the APF method create real-time paths to avoid these obstacles (Safeea et al., 2019). The effectiveness of these real-time path generation strategies depends on algorithm's immediate responsiveness, sensor accuracy, and control system precision.

Nevertheless, due to unpredictability of human movements, pre-collision control measures may not fully eliminate the risk of harmful interactions (Hjorth & Chrysostomou, 2022). Consequently, there is a growing focus on research into collision detection and post-collision feedback (Hjorth & Chrysostomou, 2022). The collision detection system relies on monitoring and predicting the robot's status using advanced methods such as CNN and SVM (Heo et al., 2019; K. M. Park, Kim, Park, & Park, 2020).

Compliance control emerges as an important aspect of HRC. Unlike traditional robots, which are constrained by "hard" programming and lack feedback control capabilities during collisions, robots engaged in HRC disassembly must adapt to external variables such as force, displacement, and rotation (Buerger & Hogan, 2007), aiming for both stability and compliance in control (Villani & De Schutter, 2016).

Direct force control, when the end-effector is in contact with the environment, is used to regulate the error between internal and external forces to achieve compliance control (Hjorth & Chrysostomou, 2022). Impedance control, representing a form of indirect force control, offer a more stable interaction mode by calculating displacement due to external changes and generating a compensatory reaction force.

Research has shown that spatially variable impedance control strategies, tailored for kinematically redundant manipulators commonly used in HRC systems, excel over static impedance control methods.

Techniques such as Cartesian impedance control and improvements in spatial redundancy resolution have been shown to enhance manipulator compliance (Ficuciello, Villani, & Siciliano, 2015).

An energy-based control strategy presents another solution for addressing spatial impedance control challenges within HRC systems. Here, the interaction between humans and robots is constrained by energy and power limits, but energy reserves can confer passivity on the system, thus stabilising impedance control within collaborative robot (Raiola, Cardenas, Tadele, De Vries, & Stramigioli, 2018). The study shows that this approach is effective in resolving the imbalance state due to disassembly uncertainty (Hjorth, Lamon, Chrysostomou, & Ajoudani, 2023).

To augment robot compliance, many systems incorporate robots with soft joints (Calanca, Muradore, & Fiorini, 2015). Robots equipped with joint torque sensors, like the KUKA iiwa LBR robots, exhibit notable performance in impedance control. Nonetheless, issues such as active control adjustments, computational delay, and non-configured delay time may impact the stability of impedance control systems in practice (Berret, Ivaldi, Nori, & Sandini, 2011).

Admittance control, another indirect force control approach, adjusts position based on force fluctuations (Keemink, van der Kooij, & Stienen, 2018). This method shows less reliance on precise environmental modelling compared to impedance control, and can better accommodate certain dynamic environmental challenges like friction (M. Schumacher et al., 2019). Efforts to merge impedance and admittance control have led to hybrid strategies that blend the benefits of both, enhancing HRC system stability and efficiency (Ott, Mukherjee, & Nakamura, 2010).

Compliant robotic control strategies facilitate direct human interaction with robots during disassembly, leveraging force sensors to interpret human intentions. This foundational force control approach has been applied across various domains, including transportation, object manipulation, and support in maintaining posture (Ikemoto, Amor, Minato, Jung, & Ishiguro, 2012).

However, the mutual cognition between robots and humans is still a challenge for HRC, including information such as materials, environments, and methods (S. Li et al., 2023), which is important in the disassembly of EV batteries. For example, distinguishing forces in the complex and unpredictable environments of HRC EV battery disassembly poses a significant challenge, especially in separating environmental factors from human active control.

3.3.6. Teleoperation techniques

Another method to enhance safety in EV battery disassembly involves teleoperation, which has seen effective use in dangerous environments like bomb disposal (Day, Bethel, Murphy, & Burke, 2008). The core concept behind teleoperation is facilitating non-contact information exchange between humans and robots. This can be accomplished through motion tracking with wearable devices or sensors, allowing for the remote control and manipulation of robots to carry out disassembly

tasks (Moniruzzaman et al., 2021). Direct control requires an operator to closely control robot motions and actions. In supervisory control, an operator's role is more to provide guidance and makes certain decisions, rather than controlling actions in every step. Multimodal control, the most advanced method, leverages AI to support decision making and motion planning processes and thus resulting in more reliable and efficient teleoperation. Table 6 outlines various teleoperation control techniques.

In these approaches, the human operator employs a communication link to oversee and guide the robot's action while gathering data about the distant environment (Moniruzzaman et al., 2021). Nonetheless, remote control operations are challenged by issues such as command delays, which come from the time taken to transmit information (MacKenzie & Ware, 1993). To counteract the impacts of uncertainties, time delays, and external interferences, ML techniques have been integrated with traditional control methods in remote operation system (Zhijun Li & Su, 2013).

Teleoperation studies of disassembly have already been published (Schmidt, Kron, & Hoogen, 2003). Teleoperation is seen as an important solution due to the serious safety risks associated with EV batteries (Hathaway et al., 2023). Some teleoperation techniques have been attempted for EV battery disassembly tasks. Mixed reality (MR) based disassembly operating platforms have been applied for EV battery pack disassembly (Rastegarpanah et al., 2024). Researchers have also applied Virtual Reality (VR) in RD of EV battery (F. Zhao & Pham, 2023).

Effective communication between humans and robots is essential for teleoperation. To facilitate smooth interaction, enhancing the capability for information exchange is vital. This goal can be accomplished by integrating a communication system with the perception system, enabling robots to interpret human intentions on a physical level (Klingspor, Demiris, & Kaiser, 1997). Fig. 8 shows some teleoperation control programs designed to support this objective. Fig. 8(a) illustrates the framework for the application of unmanned aerial vehicle (UAV). The remote-control process, depicted in Fig. 8(b), enables the local end to transmit motion commands to the remote end as needed, allowing the remote end to perform tasks autonomously. During this process, the local end retains control over the remote end. This is exemplified by a Baxter robot executing a hot-stabbing motion. The framework in Fig. 8(c) is deployed on the UR robot and performs an engine assembly task.

Visual control is a common and effective approach in remote control. Robots utilise vision sensors, such as RGB-D cameras, inertial measurement unit (IMU) sensors, or infrared imaging, to identify and interpret human gestures, including movements of the hands, heads, limbs, and even eye gaze (Kumičáková, Rengevič, Císar, & Tlach, 2017; Zhe Li, Hofemann, Fritsch, & Sagerer, 2005). ML methods, including SVM (S. Sharma, Modi, Rana, & Bhattacharya, 2018), Hidden Markov Model (HMM) (Berg, Reckordt, Richter, & Reinhardt, 2018), and Gaussian Mixture Model (GMM) (Piyathilaka & Kodagoda, 2013), have been extensively applied for gesture recognition. For example, a CNN

Table 6
Common teleoperation methods for human-robot collaboration.

Method	Features	Techniques	Ref.
Direct teleoperation	• Rely on visual feedback. • Based on the principle of direct teleoperation.	• Radio frequency (RF) and Bluetooth communications • Remote control interface • Traditional controllers such as joysticks • Visual detection	(Hathaway et al., 2023; Odesanmi, Wang, & Mai, 2023)
Supervisory teleoperation	• The operator and the robot being operated share the responsibility and control of the overall system. • Operators provide more adopted decision making for batch control.	• Object Detection and Tracking • Hierarchical Control Technology	(Muszynski, Stückler, & Behnke, 2012)
Multimodal teleoperation control	• Collect and synthesise multiple sensors and provide teleoperators with a multimodal view. • Efficient control command generation tools and good situational awareness can be provided.	• Visual and non-visual (i.e. force, sound and bio-signal) feedback • Wearable gloves and gesture-based feedback • Sensory augmentation technologies (i.e. virtual reality, augmented reality, mixed reality)	(Marino et al., 2020)

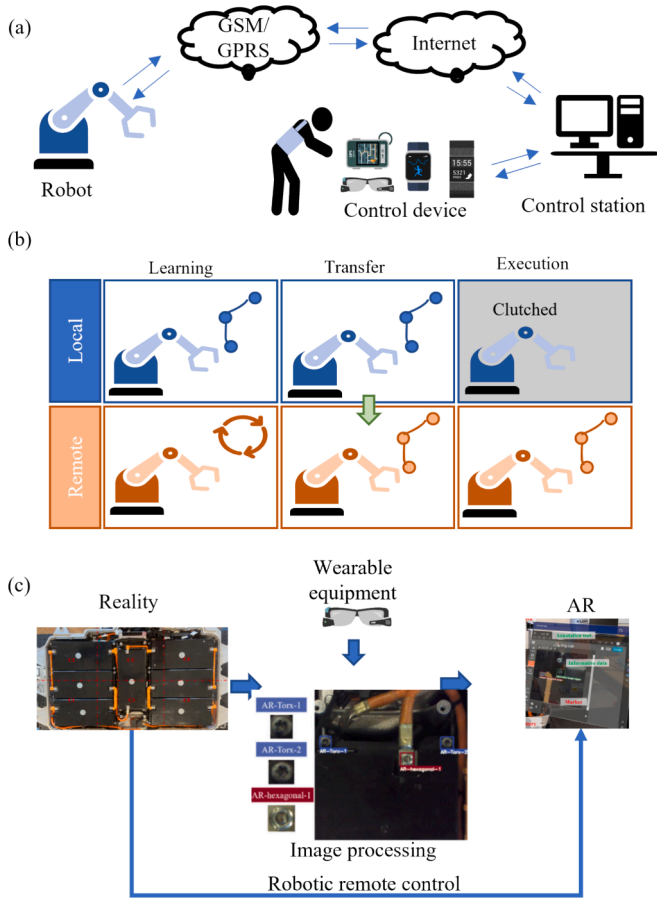


Fig. 8. Three typical frameworks of robotic remote control. (a) A robotic remote control framework by Global System for Mobile communication (GSM) (Solidakis et al., 2017). (b) A supervisory teleoperation process (Havoutis & Calinon, 2017). (c) An application of EV battery disassembly with augmented reality (Deng et al., 2024).

model based on a supervised algorithm further improves the recognition accuracy and is able to reach more than 90 % (Ji, Xu, Yang, & Yu, 2012).

Employing biological signals, such as electromyography (EMG) or electroencephalography (EEG) signals to predict human intentions presents a promising approach within HRC systems. This approach finds broad applications in imitating human behaviour, evident in technologies like prosthetics (Jiang, Englehart, & Parker, 2008) and exoskeletons (Rosen, Brand, Fuchs, & Arcan, 2001). These biosignals can predict operator conditions like fatigue or anxiety (Rani, Sarkar, Smith, & Kirby, 2004) and enable robots to synchronise with human movements.

Voice control constitutes another vital strategy. Speech recognition algorithms facilitate commands to pause, stop, or continue the execution of collaborative tasks (Bingol & Aydogmus, 2020), offering an intuitive means for humans to interact with robots.

However, the efficacy of these human-robot communication methods can be compromised by the complexities of the industrial environment. To address these challenges, integrating multiple control modes has been suggested (Maurtua et al., 2017). A multimodal interaction method can be achieved through a comprehensive robot information system framework. Multimodal remote-control systems have been proposed for robotic hand, incorporating visual, auditory, and haptic systems (Triantafyllidis, Mcgreavy, Gu, & Li, 2020). Researchers generally agree that this integration enhances operational performance.

Augmented Reality (AR) offers another technique to augment environmental perception through the human visual system. It facilitates a pre-disassembly understanding of physical objects within the shared space, enhancing awareness (Green et al., 2009). VR, in contrast, is

geared more towards simulating robot motion and exploring robot action prediction in HRC, thus boosting the operator's confidence and sense of safety (Matsas, Vosniakos, & Batras, 2018). MR technology proves beneficial for teleoperation tasks in real-world with spatial constraints, with innovations such as stereo vision and virtual grippers enabling flexible robotic manipulation (F. Chen et al., 2016).

3.4. Robotic manipulation for RD of EV battery

The robotic manipulation of EV battery disassembly raises two critical questions: 'WHICH machines and tools are suitable for disassembly operations?' and 'HOW can these machines and tools be effectively utilised and controlled?' (Vongbunyoung & Chen, 2015).

Robots excel in grasping manipulating objects within repetitive and familiar environments (Billard & Kragic, 2019). The current suite of robotic tools for disassembly is optimised for managing simple, rigid objects and executing specific disassembly tasks, such as removing separated objects or loosening screws. Addressing the requirement for robots to handle flexible materials and complex structures in RD indicates a move towards equipping robots with more dexterous hands (Billard & Kragic, 2019).

The disassembly process executed by robots is mainly driven by robot programming. Current robotic systems largely depend on offline programming (Heimann & Guhl, 2020), which typically involves the use of a compiler. Such a compiler is expected to incorporate fundamental aspects of robot motions (e.g. point-to-point motion) and may also integrate sensor feedback (e.g. force and torque). There are two principal types of programming interfaces: graph-based programming and text-based programming (Heimann & Guhl, 2020). Graph-based programming is the simplest way for someone with limited programming knowledge to get started. It involves dragging and dropping elements instead of writing code while still requiring the use of coding logic. Text-based coding involves typing instructions in a programming language following a syntax, which is the grammar and spelling rules of a programming language.

A critical focus is enhancing the flexibility of robot programming. Lead-through programming, where robots are programmed through hand-guiding (Fischer et al., 2016), serves as one such innovation. Learning-from-demonstration represents another method, capturing human motions and replicating them in robots (Zhu & Hu, 2018).

As AI progress, the intelligence of robots is set to increase. This advancement holds the promise of freeing robots from the confines of conventional programming techniques. In complex and flexible scenarios, such as RD, robots could autonomously learn and acquire disassembly skills (Siciliano, Khatib, & Kröger, 2008). This approach enhances and broadens robotic intelligence by utilising sensors and actuators to process environmental inputs and execute corresponding actions. Robots use sensors to monitor their environment, while computers analyse the data to produce commands for the robot controller, which in turn operates the robot manipulator (Z. Liu et al., 2022).

This approach diverges from traditional robotic manipulation by not predefining specific skills for the robot; instead, it allows the robot to learn skills through algorithms. Learning algorithms enable robots to acquire skills from a variety of data source and procedures.

Overall, equipping disassembly tools with more advanced robotic learning algorithms can significantly enhance the efficiency of RD for EV batteries. This section will focus on reviewing basic disassembly tools alongside more dexterous manipulators and explore robotic manipulation techniques, emphasising robotic learning.

3.4.1. Disassembly tools

Given the complexity of EV battery disassembly components, a diverse array of end effectors is required. A robotic gripper stands out as an essential tool for this purpose (Birglen & Schlicht, 2018). As shown in Fig. 9, the two-finger gripper and suction gripper emerge as the common types of grippers employed in EV battery disassembly. Both types of

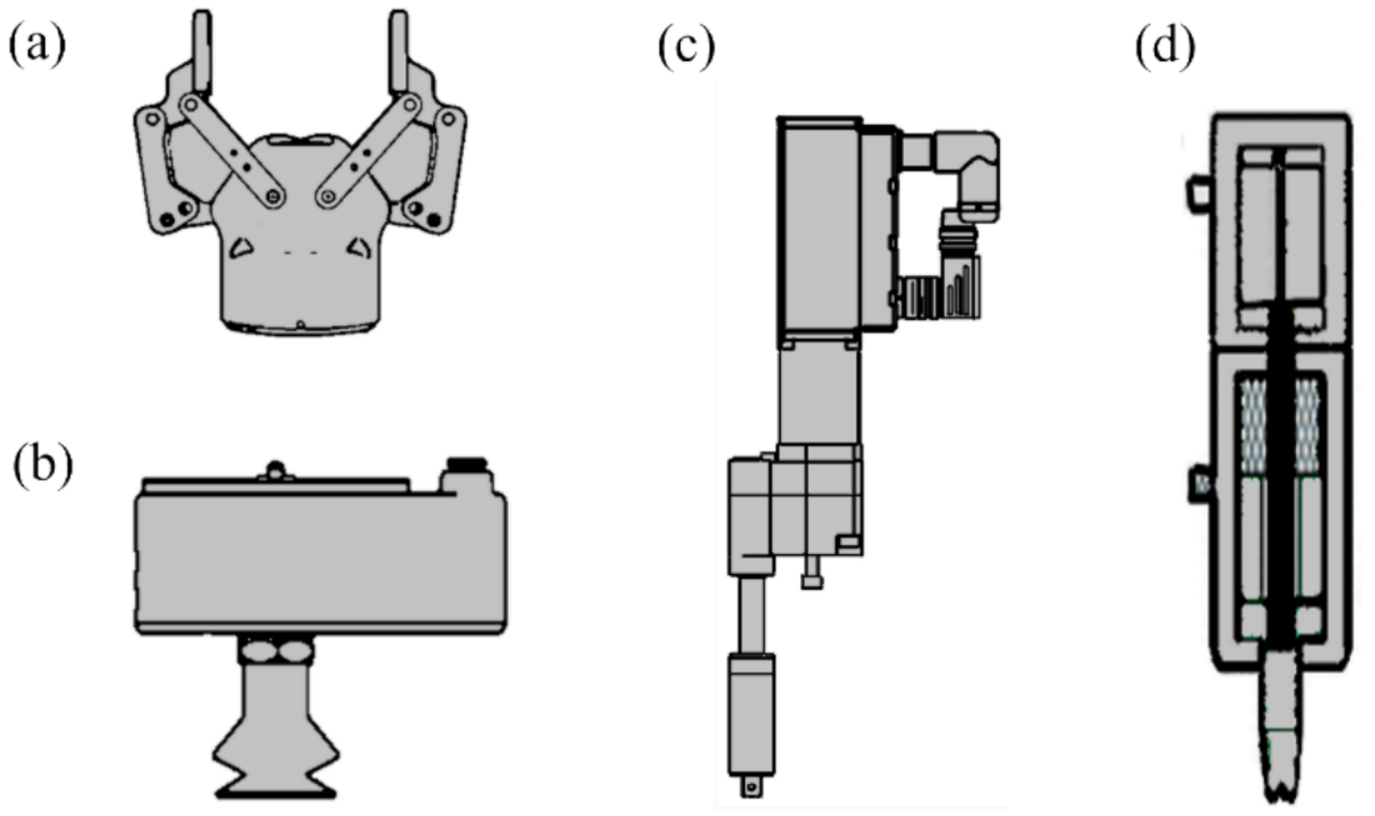


Fig. 9. Different types of grippers. (a) Two-finger gripper and (b) Vacuum gripper are mentioned in an EV battery disassembly workstation (Qu et al., 2024). (c) and (d) are automatic unscrewing tools (DiFilippo & Jouaneh, 2017; R. Li et al., 2020; Seliger, Keil, Rebafka, & Stenzel, 2001).

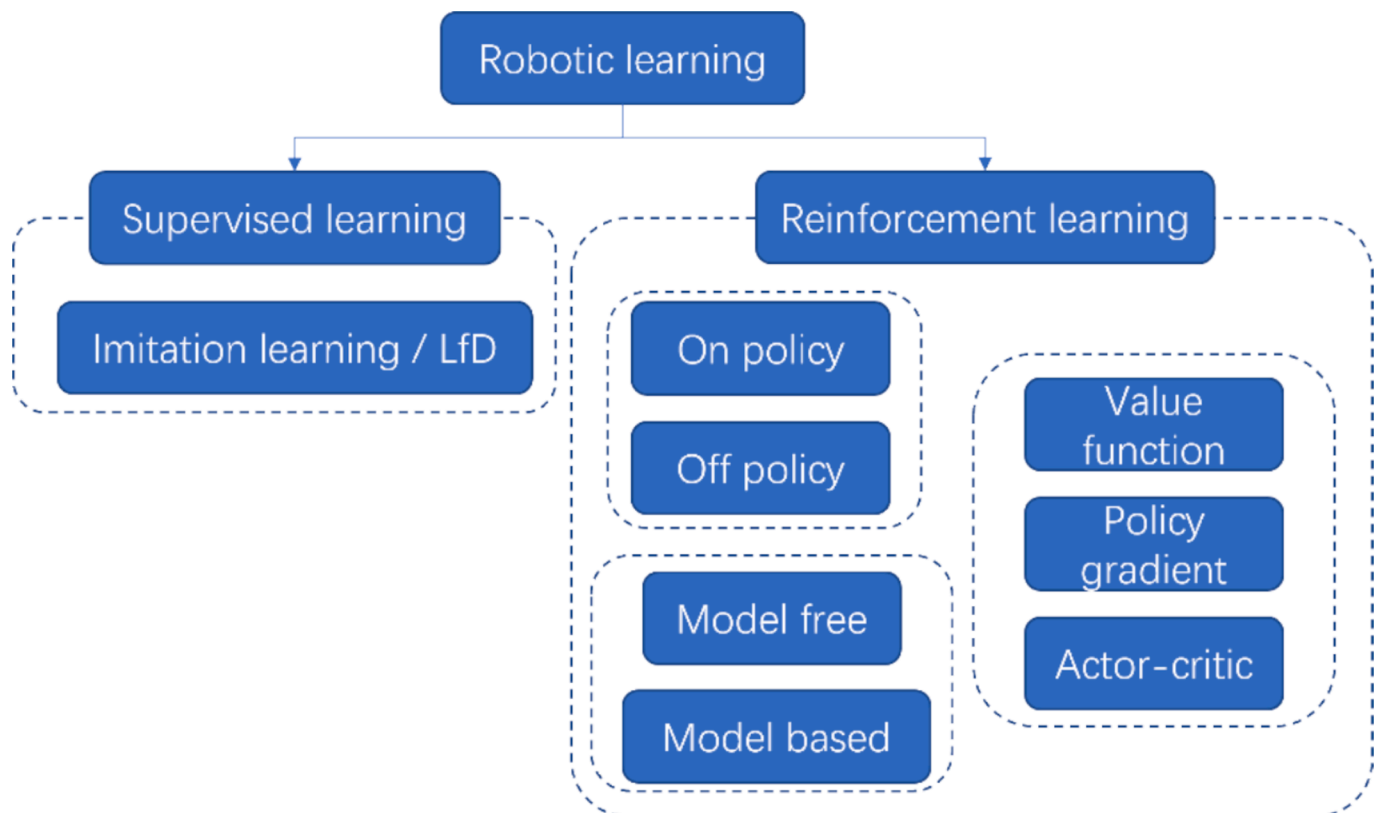


Fig. 10. The classification of robotic learning (Z. Liu et al., 2022).

grippers are adaptable, capable of managing object with varying materials, shapes, and sizes. Typically, two-finger grippers are utilised for smaller items, such as screws, bolts, nuts, and bars, whereas suction grippers are better suited for larger elements, like casing and battery modules.

The common operation in EV battery disassembly involves unfastening, which includes the removal of threaded components such as screws (Laili et al.). EV batteries feature a variety of screws, prompting the development of many screw disassembly tools. An example from of such a tool is illustrated in Fig. 9d, as presented by Seliger et al. (Seliger et al., 2001). M.DiFilippo et al. (DiFilippo & Jouaneh, 2017) developed a screw disassembly tool for laptops, equipped with vision and force sensing capabilities. Additionally, Li et al. (R. Li et al., 2020) outlined a non-destructive unfastening control strategy for unfastening hexagonal headed screws, which includes a search process to address tool-screw misalignment, as shown in Fig. 9c.

Beyond screws, snap-fit represent another key aspect of EV battery disassembly, for which specialised tools has been developed (P. Schumacher & Jouaneh, 2013).

The diversity of disassembly tools underscores the necessity for developing tool-changing systems. A significant engineering challenge within such systems is devising a mechanism that ensure high precision and reliability (Little, 2003).

Moreover, the adoption of more dexterous robotic manipulators offers a broader solution to the challenges of robotic manipulation in disassembly (Billard & Kragic, 2019). As shown in Fig. 10, these advanced manipulators are used for grasping or manipulating various objects, facilitating a more adaptable approach to RD. While anthropomorphic robots provide inspiration, their complex mechanism designs and rigid movements pose practical constraints. However, compliance control techniques can simulate the adaptability of a 'soft hand' by compensating contact forces (Y. Li & Kao, 2001). The incorporation of flexible materials into robotics, leveraging pneumatic, hydraulic and other power mechanisms, enable more versatile manipulation (Deimel & Brock, 2016).

Furthermore, the concept of multimodal robotic hand designs has been introduced (F. Liu et al., 2023). In RD, the integration of manipulator control within the comprehensive sensing system significantly enhances the RD capabilities. Particularly, leveraging both visual and haptic feedback for the manipulation of irregularly shaped objects broadens the capability of disassembly operations.

3.4.2. Development of ML technologies

ML has emerged as an important tool in robotics research, significantly enhancing RD techniques. ML encompasses various domains, including signal processing, algorithm development, and control optimisation, with the primary goal of training function using data to apply them to new situations.

Feature engineering, a critical phase in the ML workflow, entails extracting important features from data. Various models, such as linear models, quadratic models, and GMM, are employed to present the trained function. Deep neural networks have been gained popularity over these traditional models due to their proficiency in managing higher dimensional data, such as images (Z. Liu et al., 2022).

The learning stage in ML focuses on estimating the parameters within a selected model based on the provided data. This stage is categorised into three main subtypes: unsupervised learning, supervised learning, and RL. RL is deemed particularly suitable for robot learning, as it facilitates skill acquisition through experience and maximisation of rewards (Silver, Singh, Precup, & Sutton, 2021). However, RL faces challenges with high-dimensional data. Deep reinforcement learning (DRL) addresses this limitation by enhancing RL's capability to process high-dimensional data, making it a strong candidate for robot learning application (Z. Liu et al., 2022).

Robotic learning methods can be distinguished by their underlying algorithms, as illustrated in Fig. 10. These methods divide into imitation

learning and RL. Imitation learning, or Learning from Demonstration (LfD), utilises supervised learning with historical data, whereas RL operates independently of expert data and is subdivided into model-based and model-free learning approaches. Further, RL is categorised by algorithm type, such as on-policy and off-policy learning methods.

3.4.3. Robotic learning for manipulation

RD is a complex task characterised by many contact-rich challenges. To effectively operate in unstructured environments, robots require the use of high-precision sensors and a closed-loop control system (Z. Liu et al., 2022). Specifically, for EV battery disassembly, robot must possess the capability to execute movements, grasp, cut, and unscrew in complex environments. However, automating this process with traditional industrial robots poses significant challenges due to the complexity and uncertainty related to the geometry, mass, and operational environment of the objects involved.

Robotic learning methods have demonstrated potential in some assembly cases, such as steam cooker (Akkaladevi, Plasch, Pichler, & Ikeda, 2019), toy car (Duque, Prieto, & Hoyos, 2019) and valve body (Hong, Liu, Cheng, & Chen, 2018). Table 7 presents some of the robotic manipulation research solutions achieved by using robotic learning methods.

Grasping is a critical task in EV battery disassembly, where the accurate grasping of objects is essential (de Souza et al., 2021). From a robotic learning standpoint, the primary challenge in grasping is object selection and action realisation. Therefore, most grasping actions trained using robotic learning are based on image information. Researchers have used DRL or deep neural networks, such as CNN, to process images and achieve grasping ability (Berscheid et al., 2021; Moosmann et al., 2021). This method has a high success rate for robot grasping (Meng et al., 2022), but its application in complex industrial scenarios remains a challenge.

Another important development is deformable component grasping, which involves handling and removing cables and wires from EV batteries. One study implemented deformable object removal through image learning based on an inverse dynamics model (Nair et al., 2017). However, EV battery disassembly involves rich contacts between deformable objects and rigid objects, which has not been sufficiently studied in the literature.

Disassembly is a new task for robotic learning. The researchers build a RL framework to execute the unscrew task in a simulation (Kristensen et al., 2019). A door-chain disassembly task was demonstrated in real-world RL (Qu, Wang, & Pham, 2023). EV batteries have many snap fits, such as network cable ports, charging ports and coolant ports. Currently, there are no reports of AI methods being applied in the industry for EV battery disassembly. This is not only due to the complexity and non-structural nature of the disassembly action but also the technical readiness level of AI methods in the industry.

4. Discussion: Potential assistant technologies of robots to EV battery disassembly

This section discusses potential technologies that can impact the use of RD in EV batteries. These assistive technologies are crucial for enhancing the feasibility of using robots in EV battery disassembly.

4.1. Optimisation of destructive/semi-destructive disassembly processes in RD

Implementing non-destructive disassembly can be costly, unreliable and slow due to the complexity of sensor and actuator systems involved. As a result, opting for destructive disassembly or semi-destructive disassembly might present a more viable alternative (Herrmann et al., 2012). However, these methods tend to increase safety risks.

One example is the adoption of robotic cutting tool over traditional manual cutting in EV battery disassembly (Kay et al., 2022). Another

Table 7

Some industrial applications based on learning methods.

Field	Ref.	Task	Methods	Advantages	Disadvantages
Robotic grasping or handing	(Mahler et al., 2019)	Ambidextrous robot grasping	Supervised Learning with Neural Networks Learning	• The trained model has good portability.	• A large amount of real-world data is required for training.
	(de Souza, Rocha, Oliveira, Moreira, & Boaventura-Cunha, 2021)	Robotic grasping policy			
Robotic assembly	(Berscheid, Friedrich, & Kröger, 2021)	Pick and place			
	(Mohammed, Chung, & Chyi, 2020)	Pick and place in cluttered scene	RL	• No need for pre-training	• The efficiency of objective recognition is low.
Robotic assembly	(Moosmann et al., 2021)	Separate entangled workpieces			
	(Y. Wang et al., 2021)	Grasp for assembly			
Robotic assembly	(Duque et al., 2019)	Toy car	Imitation Learning	• No need to manually set the model or reward function.	• High quality and adequate assembly demonstration are required.
	(Hong et al., 2018)	Valve body			
Other robotic process control	(Cheng, Chen, Hao, & Li, 2014)	Transmission part		• Acquire assembly skills faster.	
	(Akkaladevi et al., 2019)	Steam cooker	RL	• No need for tedious assembly demo collections.	• A large gap between simulation and reality.
Other robotic process control	(J. Luo, Solowjow, Wen, Ojea, & Agogino, 2018)	Insert peg into deformable hole			
	(J. Luo et al., 2019)	Gear			
Other robotic process control	(Mueller, Venicx, & Hayes, 2018)	Pouring	Imitation Learning	• Capable of adapting to more complex robotic processes.	• Limited generalisation ability
	(Nair et al., 2017)	Robotic rope manipulation			
Other robotic process control	(Ying Zhang, Li, & Yang, 2021)	Robotic ultrasound scanning			
	(Fu, Li, Zheng, & Song, 2021)	Robotic fabric manipulation	RL	• It is more suitable for various planning tasks in robots	• The amount of computation in the process of generating the results
Other robotic process control	(Maldonado-Ramirez, Rios-Cabrera, & Lopez-Juarez, 2021)	Visual path-following in welding			• Completely dependent on data
	(Sichkar, 2019)	Path planning for mobile robot			

publication proposes a tool to automate the separation of joints (Tan et al., 2021). Overall, this research on automation of destructive disassembly is lacking.

Semi-destructive disassembly techniques, such as shearing, pulling, and prying, are typically employed at the module level. These methods necessitate precise force control and advanced dexterity. A design method using split lines has been proposed and applied to heat exchangers to improve the efficiency of semi-destructive disassembly (Umeda, Miyaji, Shiraishi, & Fukushige, 2015). However, a noticeable development gap exists in automating these semi-destructive tasks.

In summary, there is a notable scarcity of research into achieving an optimal balance among non-destructive, semi-destructive and destructive disassembly operations to maximise safety, cost-efficiency and operation effectiveness.

4.2. Design for disassembly

The design of EV batteries can result in very different disassembly processes and thus disassembly difficulty needs to be considered in the design phase. Design for disassembly (DFD) aims to reduce the complexity of EV battery disassembly while also considering assembly requirements (Meng et al., 2022). DFD has the potential to save resources, energy, and costs, and promote high cyclability in EV batteries.

4.3. Technology readiness level of AI-based techniques for EV battery disassembly

There have been numerous industrial trials and lab-scale prototypes for the RD of EV batteries, particularly using schedule-based methods that have been partially implemented in the industry (Wegener et al., 2014). However, the use of AI methods to create smart disassembly is still new to the industry (Meng et al., 2022). A significant number of AI methods are involved in HRC disassembly and schedule-free methods.

From the perspective of technology readiness level (Meng et al., 2022), AI-based intelligent disassembly of EV batteries is still in the early stages and has yet to be verified in real industrial scenarios.

5. Conclusion

This article has presented an extensive review of the RD of EV batteries. The contribution of this work can be summarised in three aspects:

First, it identifies and discusses the challenges in the RD of EV batteries. Battery design, dismantling tasks and dismantling objectives vary widely, and EoL EV batteries carry a great deal of uncertainty and safety risk. This requires RD systems to be more efficient and flexible.

Second, this paper summarises research on the technologies related to RD and presents key technologies for the RD of EV batteries. The review finds that the RD of EV batteries is easier to achieve using pre-programmed motions. However, the RD of EV batteries still needs improvement in terms of automation and flexibility. There are already a number of techniques in robotic inspection, planning, and manipulation that can be applied to EV battery disassembly.

Third, this paper discusses assistive technologies in the RD of EV batteries. These technologies are essential for expanding the use of robots in EV battery disassembly.

CRedit authorship contribution statement

Yue Zang: Writing – original draft, Methodology, Investigation, Conceptualization. **Mo Qu:** Writing – review & editing, Methodology. **Duc Truong Pham:** Writing – review & editing. **Roger Dixon:** Writing – review & editing. **Farzaneh Goli:** Writing – review & editing. **Yongquan Zhang:** Writing – review & editing, Conceptualization. **Yongjing Wang:** Writing – review & editing, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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